

QUIZAPP

Java Mini Project Report

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Project	QuizApp
Technology	Java
Application Type	Console-based Quiz Application

This report documents the purpose, scope, implementation concepts, and working flow of the Java QuizApp mini project.

1. Project Overview

QuizApp is a Java console-based quiz application that allows the user to select a difficulty level—Easy, Medium, or Hard—and answer three randomly selected questions from that level. The application checks each answer, maintains a score, and displays the final score and percentage.

2. Objectives

- Develop a simple interactive quiz application using Java.
- Store quiz questions, options, difficulty levels, and correct answers in a structured two-dimensional array.
- Allow the user to choose a difficulty level using switch-case.
- Randomly select questions while avoiding repeated questions.
- Accept and validate user answers using String operations.
- Maintain the user's score and calculate the final percentage.
- Demonstrate core Java programming concepts in a practical mini project.

3. Scope

The current scope of QuizApp is a console-based Java quiz with three difficulty levels and three questions per attempt. Questions and answers are stored directly in the program, and the application runs locally through the Java console.

- Difficulty selection: Easy, Medium, and Hard.
- Three questions are asked for the selected difficulty.
- Questions are selected randomly.
- The same question is not selected twice in one quiz attempt.
- The user answers using A, B, or C.
- The final score and percentage are displayed after the quiz.

4. Concepts Used

Concept	Use in QuizApp
Class	QuizApp is the main class containing the application logic.
Objects	Scanner, Random, and ArrayList are used as objects created from Java classes.
Scanner	Reads the user's input from the console.
Two-Dimensional Array	String[][] q stores the difficulty, question, three options, and correct answer for each question.
Variables and Data Types	int is used for numeric values such as choice and score, while String is used for text.
Switch-Case	Maps the user's numeric choice to Easy, Medium, or Hard.
If-Else	Checks whether the entered answer matches the correct answer.
For Loop	Repeats the question process three times.

Concept	Use in QuizApp
Do-While Loop	Generates a random question index and repeats until the difficulty matches and the question has not been used.
Random	Generates random question indexes.
ArrayList	Stores indexes of already selected questions to prevent repetition.
Methods	The quiz() method separates quiz processing from main() and returns the score.
Parameter and Return Value	String level passes the selected difficulty to quiz(), while return score sends the final score back to main().
String Methods	toUpperCase() normalizes the user's answer, and equals() compares String contents.
Operators	Increment, logical, arithmetic, and comparison operations are used throughout the program.
Static	The Scanner and quiz-related members are accessed at class level without creating a QuizApp object.

5. Working of the Application

Step 1 — Start: The program starts from the main() method and displays the quiz title and difficulty options.

Step 2 — Select Difficulty: The user enters 1, 2, or 3. A switch-case converts this choice into Easy, Medium, or Hard.

Step 3 — Start Quiz: The selected difficulty is passed to quiz(difficulty).

Step 4 — Select Questions: A Random object generates indexes. A do-while condition ensures the question has the selected difficulty and has not already been used.

Step 5 — Display Question: The selected row of the 2D array is used to display the question and its three options.

Step 6 — Check Answer: The entered answer is converted to uppercase and compared with the stored correct answer using equals().

Step 7 — Update Score: For every correct answer, score++ increases the score by one.

Step 8 — Display Result: After three questions, quiz() returns the score. main() displays the score and calculates the percentage.

6. Question Data Structure

Each question is stored as one row of the two-dimensional array q. The six columns follow this structure:

Index	0	1	2	3	4	5
Stores	Difficulty	Question	Option A	Option B	Option C	Correct Answer

7. Key Code Logic

The core question-selection condition is designed to reject questions that do not match the selected difficulty or that were already used:

```
while (!q[n][0].equals(level) || used.contains(n));
```

The percentage is calculated using the score out of three questions:

```
(score * 100 / 3.0)
```

8. Expected Output

The application displays the selected difficulty, three questions with options A, B, and C, feedback for each answer, and a final result containing the score and percentage.

9. Conclusion

QuizApp demonstrates how fundamental Java concepts can be combined to create a functional interactive application. It uses arrays, collections, control statements, methods, objects, randomization, String handling, and basic calculations to implement the complete quiz flow.

10. Project Repository

This report is prepared as supporting documentation for the QuizApp Java mini project and can be included with the project source code in the GitHub repository.